Ames Housing Price Prediction

Final report: data, insights, models and final selection

Ames, Iowa dataset | Target: SalePrice

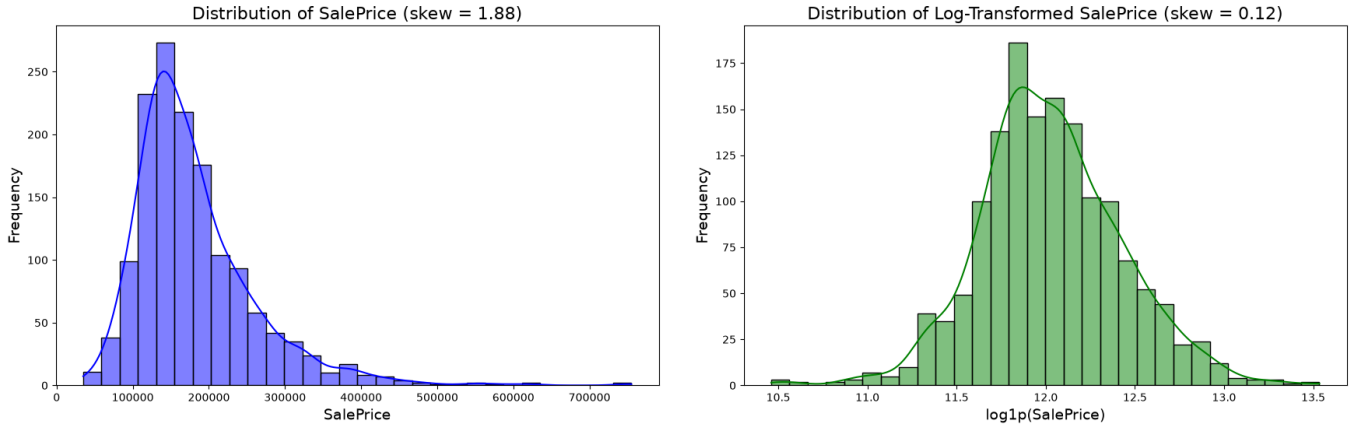
1,460	81	$180,921	0.943
HOUSES (OBSERVATIONS)	TOTAL ATTRIBUTES	MEAN SALE PRICE	TEST R^2 (TUNED SVR)

1. Dataset & Preprocessing

- **Data:** 1,460 houses x 81 columns, target `SalePrice` (mean $180,921, median $163,000, skew 1.88). No duplicate rows. Feature types: 35 numerical, 26 nominal and 18 ordinal (with explicit order, e.g. Po < Fa < TA < Gd < Ex).
- **Missing values:** in 15 columns (basement, garage, pool, fence, alley, ...) NA means "feature absent", so it became the category `'None'`. Real missing values remain in 4 columns: LotFrontage (17.74%), GarageYrBlt (5.55%), MasVnrArea (0.55%) and Electrical (0.07%); they are imputed inside the pipeline (median for numerical, most frequent for categorical).
- **Cleaning and features:** two outliers were removed (Id 524 and 1299: 4,676 and 5,642 sq ft but only $184,750 and $160,000, both Partial sales), leaving 1,458 houses. 8 features were engineered (TotalSF, TotalBath, HouseAge, RemodAge, Qual_x_Area, HasGarage, HasBasement, HasPool), giving 87 features (43 numerical, 26 nominal, 18 ordinal).
- **Transformations:** the target uses `log1p` (skew 1.88 to 0.12; predictions are converted back with expm1, so all metrics are in dollars). 21 skewed numerical features (|skew| > 0.75, found on the training set) use log1p for the SVR and Yeo-Johnson for the other models, followed by `RobustScaler` (median/IQR, not affected by the remaining outliers; SVR needs scaled inputs).
- **Pipelines:** three pipelines combined with a ColumnTransformer: numerical (median imputer, skew transform, scaler), nominal (most-frequent imputer, One-Hot: 26 columns become 186) and ordinal (most-frequent imputer, Ordinal Encoding). The output has 247 features and no NaN. Everything is fitted on the training data only (no leakage).
- **Split:** 70% train / 15% validation / 15% test = 1,020 / 219 / 219 houses, stratified by price quintile, `random_state=42`.

2. Key EDA Findings

- **Target:** SalePrice is right-skewed (1.88); after log1p the skew is 0.12, so the log target suits regression models.
- **Strongest correlations with price:** OverallQual 0.79, GrLivArea 0.71, GarageCars 0.64, GarageArea 0.62, TotalBsmtSF 0.61 (engineered: Qual_x_Area 0.87, TotalSF 0.83).
- **Multicollinearity:** GarageCars/GarageArea 0.88, YearBuilt/GarageYrBlt 0.83, GrLivArea/TotRmsAbvGrd 0.83, TotalBsmtSF/1stFlrSF 0.82. This motivates regularized models and permutation importance.
- **Skewed features:** MiscVal (24.5), PoolArea (14.8) and LotArea (12.2) are the most skewed; GrLivArea (1.37) and TotalBsmtSF (1.52) are moderately skewed.
- **Outliers:** the IQR rule flags 61 SalePrice, 31 GrLivArea, 61 TotalBsmtSF, 69 LotArea and 21 GarageArea values. Only Ids 524 and 1299 were removed; Ids 692 and 1183 ($755,000 and $745,000) are real expensive houses and were kept.



SalePrice before (skew 1.88) and after the log1p transformation (skew 0.12)

3. Data Insights: Answers to the Guideline Questions

Every answer below is taken from the output of the notebook code (Section 3).

Question	Answer (actual values)
Housing prices	
What is the average, median, minimum and maximum SalePrice?	Mean $180,921; median $163,000; minimum $34,900; maximum $755,000.
Which houses have the highest and lowest sale prices?	Highest: $755,000 (Id 692, 4,316 sq ft, quality 10) and $745,000 (Id 1183). Lowest: Id 496 at $34,900 (IDOTRR, 720 sq ft, built 1920); 3 of the 5 cheapest were abnormal sales.
How does house price vary with living area (GrLivArea)?	Strong positive relationship, r = 0.71: larger houses cost more. A few houses above 4,000 sq ft break the pattern (outliers).
What are the characteristics of the most expensive houses?	Top 5% vs all houses (medians): living area 2,364 vs 1,464 sq ft; OverallQual 9 vs 6 (92% have quality 8 or more); 3-car vs 2-car garage; basement 1,746 vs 992 sq ft; built 2005 vs 1973. Mostly in NridgHt (29 houses), NoRidge (15) and StoneBr (9).
Neighborhood analysis	
Which 3 neighborhoods have the highest average house prices?	NoRidge, NridgHt and StoneBr.
What is the average SalePrice in the top 3 neighborhoods?	NoRidge $335,295; NridgHt $316,271; StoneBr $310,499.

Question	Answer (actual values)
Which 3 neighborhoods have the lowest average house prices?	MeadowV, IDOTRR and BrDale.
What is the average SalePrice in the lowest 3 neighborhoods?	MeadowV $98,576; IDOTRR $100,124; BrDale $104,494.
Which neighborhood has the highest number of houses?	NAmes with 225 houses (then CollgCr 150 and OldTown 113).
Does neighborhood appear to have a strong relationship with house price?	Yes. Neighborhood explains 55% of the variance in SalePrice; the average price differs about 3.4x between MeadowV ($98,576) and NoRidge ($335,295).
Time-based analysis	
Which month has the highest average house sale price?	September (month 9): $195,683.
Which month has the lowest average house sale price?	April (month 4): $171,503.
Which month has the highest number of house sales?	June, with 253 houses sold.
Does the average house price vary throughout the year?	Only slightly. The gap between the best and worst month is 13.4% of the mean and month explains just 0.7% of the price variance, so there is no real seasonal effect.
House characteristics	
Does the number of bedrooms affect house price?	Partly. Median price: 2 bedrooms $137,250; 3 bedrooms $169,945; 4 bedrooms $193,500. The 5- and 6-bedroom groups are small (21 and 7 houses) and do not keep rising ($161,500 and $141,000).
How does the number of bathrooms relate to SalePrice?	Price rises with full bathrooms. Median: $132,375 (1 bath), $196,750 (2 baths), $320,000 (3 baths, only 33 houses).
How does overall house quality (OverallQual) affect price?	Strong positive relationship, r = 0.79: median price rises from $50,150 (quality 1) to $432,390 (quality 10).
How does garage capacity affect house price?	Median price: $100,000 (0 cars), $128,000 (1), $177,750 (2), $295,000 (3). Only 5 houses have a 4-car garage ($200,000), too few to conclude.
Does having a basement significantly affect the average house price?	Yes, the difference is large: median $165,000 with a basement vs $101,800 without (+62.1%). Only 37 houses have no basement, and no formal significance test was run.
How does house age affect SalePrice?	Negative relationship, r = -0.52: newer houses cost more. HouseAge is also among the top 7 features of both final models (Section 6).
Additional analysis	
Is there a price premium for 3-car garages vs 2-car garages?	Median $295,000 (181 houses) vs $177,750 (824 houses): +$117,250 (+66.0%). These houses are also larger and of higher quality, so the premium is not due to the garage alone.

4. Models Used & Hyperparameter Tuning

Five regression models were trained as full pipelines (preprocessing, then model) and compared with a median-price baseline: SVR (required), KNN, Ridge, Lasso and ElasticNet. Linear Regression was dropped because it was numerically unstable with correlated features. Each model was tuned with **GridSearchCV** (5-fold KFold, shuffled, random_state=42; score = RMSE on the log target) using the training set only.

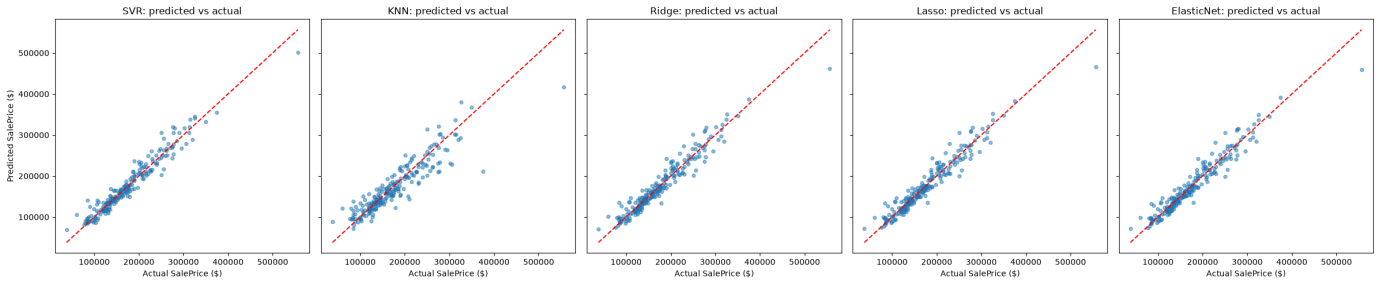
Model	Best hyperparameters	CV RMSE (log)
SVR	kernel = rbf, C = 100, gamma = 0.0003, epsilon = 0.05	0.1083
KNN	k = 5, p = 1 (Manhattan), distance weights	0.1528
Ridge	alpha = 10	0.1090
Lasso	alpha = 0.0003	0.1084
ElasticNet	alpha = 0.001, l1_ratio = 0.5	0.1080

- **C** (error penalty): CV RMSE 0.1099 at C = 1 (too simple), 0.1084 at C = 10 and 0.1083 at C = 100, so any value from 10 to 100 works. **Kernel:** RBF (0.1083) beat polynomial (0.1099). **Gamma** (reach of one training point): small values were best (0.0003: 0.1083), clearly better than "scale" (0.1099) and "auto" (0.1103). **Epsilon** (error-free tube): 0.05 was best (0.1083), the default 0.1 was worse (0.1097). **Degree** (poly kernel only): 2 (0.1099) beat 3 (0.1142), because a higher degree mostly fits noise.
- **Does tuning help?** Yes for the SVR (validation R^2 0.928 to 0.942, test R^2 0.924 to 0.943). It helped Ridge and Lasso slightly (test R^2 0.926 to 0.932 and 0.933) and did not change ElasticNet (its best parameters equal the starting ones). KNN improved on validation (0.861 to 0.882) but not on test (0.843 to 0.834).
- **Extra experiment (Optuna):** an additional Optuna search for the SVR (60 trials) reached a slightly lower CV RMSE (0.1064 vs 0.1083 with GridSearchCV) and validation R^2 (0.944 vs 0.942), but the gain is small and the search goes beyond the required GridSearchCV, so it was treated as an extra experiment. The GridSearchCV-tuned SVR remains the final model.

5. Evaluation Results

Model (tuned)	Train R^2	Val R^2	Test R^2	Test MAE ($)	Test MSE	Test RMSE ($)
SVR	0.968	0.942	0.943	11,763	2.65e8	16,286
ElasticNet	0.944	0.935	0.931	12,582	3.20e8	17,878
Lasso	0.949	0.934	0.933	12,415	3.12e8	17,669
Ridge	0.949	0.932	0.932	12,499	3.15e8	17,747
KNN	1.000	0.882	0.834	18,828	7.73e8	27,800
Baseline (median price)	-0.053	-0.055	-0.037	51,859	4.82e9	69,405

Validation RMSE: SVR $18,085, ElasticNet $19,152, Lasso $19,335, Ridge $19,584, KNN $25,825. The full table (11 models, MAE, MSE, RMSE and R^2 on train, validation and test) is in Section 8 of the notebook.



Predicted vs actual SalePrice on the test set for the five tuned models (red line = perfect prediction)

- **Which model performs best?** The tuned SVR: validation R^2 0.942 (RMSE $18,085) and test R^2 0.943 (RMSE $16,286, MAE $11,763). The three linear models are close behind (test R^2 0.931 to 0.933); KNN is the weakest.
- **Overfitting / train vs validation:** the train-validation R^2 gap is small for the SVR (0.026) and the linear models (0.009 to 0.017). The tuned KNN overfits: its train R^2 is exactly 1.000 (an artifact of distance weighting) but only 0.882 on validation and 0.834 on test.
- **Underfitting:** only the median baseline underfits (R^2 about -0.05). The SVR and linear models reach train R^2 of 0.944 to 0.968 and almost the same on validation and test; the largest errors are on very cheap and very expensive houses.
- **Generalization:** the validation-to-test change in R^2 is -0.001 for the SVR, 0.000 to 0.004 for the linear models and 0.048 for KNN, so the SVR and linear models generalize well and KNN does not.

6. Feature Importance, Final Model Selection & Conclusions

- **Most important features** (permutation importance on the validation set; ElasticNet coefficients as a cross-check): **TotalSF** is first for both models, followed by OverallQual, OverallCond, GrLivArea, HouseAge, LotArea and Neighborhood; 8 of the top 10 features are shared by the two models. This agrees with the EDA and insights: size and quality have the highest correlations with price, neighborhood explains 55% of the variance and newer houses cost more.
- **Final model: tuned SVR** (RBF, C = 100, gamma = 0.0003, epsilon = 0.05). It was chosen with validation and CV results, and the test set only confirmed it. It has the best validation score of the compared models (R^2 0.942, RMSE $18,085) and a CV RMSE practically tied with the best (0.1083 vs 0.1080 for ElasticNet), a modest train-validation gap (0.026), and a test MAE of $11,763 (7.2% of the median test price). It is more complex than the linear models (556 support vectors, no coefficients), so ElasticNet (100 of 247 coefficients, smallest gap 0.009) is the simpler alternative if interpretability matters more than accuracy.
- **Main conclusions:** (1) house price is explained mostly by total area, overall quality and condition, age, lot size and neighborhood, while the month of sale has almost no effect; (2) log-transforming the target and wrapping all preprocessing in pipelines gave models with R^2 above 0.93 on unseen data; (3) the SVR is only modestly better than the linear models (about 0.01 in test R^2), and KNN is clearly worse.
- **Limitations:** one dataset (Ames, Iowa, sales from 2006 to 2010), a single split with a small test set (219 houses), and the largest errors on the cheapest and most expensive houses. **Reproducibility:** random_state = 42 everywhere, the split check passed, refitting the final model gives identical predictions, and requirements.txt was generated from the installed library versions.